

A4 :

TFD (1,1,0)

$$x_0 = 1$$

$$x_1 = 1$$

$$x_2 = 0$$

$$X_0 = x_0 + x_1 + x_2 = 1 + 1 + 0 = \underline{2}$$

$$X_1 = x_0 + x_1 * e^{[i * ((-2 * \pi) / 3)]} + x_2 * e^{i * ((2 * \pi) / 3)} = 1 + e^{[(-2i * \pi) / 3]} + 0 * e^{[(2i * \pi) / 3]} = \underline{1 + e^{[(-2i * \pi) / 3]}}$$

$$X_2 = x_0 + x_1 * e^{[i * ((2 * \pi) / 3)]} + x_2 * e^{i * ((-2 * \pi) / 3)} = 1 + e^{[(2i * \pi) / 3]} + 0 * e^{[(-2i * \pi) / 3]} = \underline{1 + e^{[(2i * \pi) / 3]}}$$

$$\text{TFD (1,1,0)} = \begin{pmatrix} 2 \\ (1 + e^{[(-2i * \pi) / 3]}) \\ (1 + e^{[(2i * \pi) / 3]}) \end{pmatrix}$$

TFD (1,0,1)

$$x_0 = 1$$

$$x_1 = 0$$

$$x_2 = 1$$

$$X_0 = x_0 + x_1 + x_2 = 1 + 0 + 1 = \underline{2}$$

$$X_1 = x_0 + x_1 * e^{[i * ((-2 * \pi) / 3)]} + x_2 * e^{i * ((2 * \pi) / 3)} = 1 + 0 * e^{[(-2i * \pi) / 3]} + e^{[(2i * \pi) / 3]} = \underline{1 + e^{[(2i * \pi) / 3]}}$$

$$X_2 = x_0 + x_1 * e^{[i * ((2 * \pi) / 3)]} + x_2 * e^{i * ((-2 * \pi) / 3)} = 1 + 0 * e^{[(2i * \pi) / 3]} + e^{[(-2i * \pi) / 3]} = \underline{1 + e^{[(-2i * \pi) / 3]}}$$

$$\text{TFD (1,0,1)} = \begin{pmatrix} 2 \\ (1 + e^{[(2i * \pi) / 3]}) \\ (1 + e^{[(-2i * \pi) / 3]}) \end{pmatrix}$$

TFD (2,1,1)

$$x_0 = 2$$

$$x_1 = 1$$

$$x_2 = 1$$

$$X_0 = x_0 + x_1 + x_2 = 2 + 1 + 1 = \underline{4}$$

$$X_1 = x_0 + x_1 * e^{[i * ((-2 * \pi) / 3)]} + x_2 * e^{i * ((2 * \pi) / 3)} = 2 + e^{[(-2i * \pi) / 3]} + e^{[(2i * \pi) / 3]} = \underline{2 + e^{[(-2i * \pi) / 3]} + e^{[(2i * \pi) / 3]}}$$

$$X_2 = x_0 + x_1 * e^{[i * ((2 * \pi) / 3)]} + x_2 * e^{i * ((-2 * \pi) / 3)} = 2 + e^{[(2i * \pi) / 3]} + e^{[(-2i * \pi) / 3]} = \underline{2 + e^{[(2i * \pi) / 3]} + e^{[(-2i * \pi) / 3]}}$$

$$\begin{aligned} \text{TFD}(2,1,1) &= (4) \\ &\quad (2 + e^{[(-2i * \pi) / 3]} + e^{[(2i * \pi) / 3]}) \\ &\quad (2 + e^{[(2i * \pi) / 3]} + e^{[(-2i * \pi) / 3]}) \end{aligned}$$

Calculs du jour : soit $w = e^{(i\pi/3)}$

1) Calculer

$$w_0 = 1 \quad w_2 = e^{(2i\pi/3)} \quad w_3 = i^2 = -1 \quad w_{-1} = -i \quad w_{-2} = e^{(-2i\pi/3)}$$

2) écrire ces 6 nombres complexes sous forme algébrique. Que remarquez-vous ?

$$\begin{aligned} w_0 &= 1 & w_2 &= -2 + \text{racine } 3 / 2 & w_3 &= i^2 = -1 & w_{-1} &= \\ w_{-2} &= (-1 / 2) - \text{racine } 3 / 2 \end{aligned}$$

On remarque que les nombres d'indices opposés ont les mêmes valeurs.